

VIGILANCE

Protecting workers. One frame at a time.

Phase 01 · Basic Detection

Weeks 1–4

WHAT YOU'LL BUILD

- ✓ Open a video file and process frame by frame
- ✓ Detect if a person is present in the frame
- ✓ Check for hard hat and safety vest
- ✓ Draw red bounding box around violations
- ✓ Print VIOLATION DETECTED on screen

TECH STACK

Python · OpenCV · YOLOv8 pretrained model

NOTE

No alerts yet. No database. Pure visual detection.

Phase 02 · Alert System

Weeks 5–6

WHAT YOU'LL BUILD

- ✓ Audio alarm when a violation is detected
- ✓ Automatically save a screenshot of the violation
- ✓ Send WhatsApp / SMS to factory manager via Twilio
- ✓ Cooldown timer to prevent alert spam

TECH STACK

Python · OpenCV · YOLOv8 · Twilio API · playsound

NOTE

First time the system reacts to what it sees.

Phase 03 · Intelligence

Weeks 7–8

WHAT YOU'LL BUILD

- ✓ Train custom YOLOv8 model on PPE dataset from Roboflow
- ✓ Detect specific gear — helmet, vest, gloves, goggles
- ✓ Zone-based detection (only alert in danger zones)
- ✓ Log violations to SQLite database

TECH STACK

Python · OpenCV · YOLOv8 custom · Roboflow · Google Colab · SQLite

NOTE

The system now knows what it's looking for specifically.

Phase 04 · Scale

After 2 months

WHAT YOU'LL BUILD

- ✓ Live CCTV feed support across multiple cameras
- ✓ Face recognition to identify which worker violated
- ✓ Web dashboard showing live feed and violation history
- ✓ Daily compliance reports auto-emailed to management
- ✓ Repeat offender tracking and escalation
- ✓ Deploy on Raspberry Pi or Jetson Nano on-site

TECH STACK

Python · Flask · SQLite · face_recognition · edge deployment

NOTE

Turns the script into a deployable product.

Phase 05 · Product

Future

WHAT YOU'LL BUILD

- ✓ Multi-factory cloud-hosted support
- ✓ Mobile app for factory managers
- ✓ Analytics dashboard with violation heatmaps
- ✓ Integration with HR and payroll systems
- ✓ Auto-generated compliance reports for audits
- ✓ SaaS model — charge factories a monthly fee

TECH STACK

React · AWS/GCP · PostgreSQL · Docker · Mobile (React Native)

NOTE

The full business product.